

Step	Step Description	Agents Involved	Tool Calls and Parameters	Reason
1	Decompose query into steps	planning_agent	N/A	planning_agent is chosen to break down the geospatial query into actionable steps, as per its system prompt.
2	Find area polygon for Exeter	named_area	call_os_ngd (polygon_or_point=True, filename='exeter_area_polygon')	named_area is chosen because the prompt specifies it contains polygons for named areas like cities. No filters are applied initially to ensure all possible polygons for Exeter are retrieved.
3	Filter to Exeter polygon(s)	data_analysis_agent	generate_metadata_for_artifacts(['exeter_area_polygon']); code_executor (filtering for Exeter using name1_text, description, descriptiongroup columns)	data_analysis_agent is used to filter the raw named area data. name1_text, description, and descriptiongroup columns are chosen based on metadata and the need to accurately identify Exeter polygons.
4	Search for houses in Exeter area	buildings	call_os_ngd (bbox='filtered_exeter_area_polygon', filename='all_buildings_in_filtered_exeter_area')	buildings agent is chosen as per its prompt for searching buildings within a given area. The bbox parameter is set to the filtered Exeter polygon artifact.
5	Filter for houses (residential, single address)	data_analysis_agent	generate_metadata_for_artifacts(['buildings_all_buildings_in_filtered_exeter_area']); code_executor (filter: buildinguse_addresscount_residential > 0 and buildinguse_addresscount_total = 1)	The definition of a house is taken from the buildings agent's system prompt. The relevant columns are identified from the metadata.
6	Identify houses likely hard to heat	data_analysis_agent	generate_metadata_for_artifacts(['houses_in_filtered_exeter_area']); code_executor (filter: buildingage_period in pre-1999 periods OR constructionmaterial in list of poor materials)	Criteria for 'hard to heat' are based on prompt instructions and metadata: pre-1999 age periods and specific construction materials. Columns are chosen from metadata. Multiple code attempts were made due to variable naming issues, resolved by passing the correct artifact as a list.
7	Plot distribution of hard to heat houses	plotting_agent	generate_metadata_for_artifacts(['hard_to_heat_houses_in_exeter_area']); code_executor (folium map, convert datetime columns to string for serialization)	plotting_agent is chosen for visualization. Folium is used as per plotting agent's template. Conversion of datetime columns to string is required due to serialization errors with folium/GeoJSON.
8	Summarize and communicate results	host_agent	N/A	host_agent summarizes the results and offers further analysis or visualization, as per its role.

Overall Summary: - Databases used: Named area (for Exeter polygon), buildings (for all buildings in Exeter), and data analysis/plotting agents for filtering and visualization. - Filters/code: Filtering for Exeter polygons used multiple columns (name1_text, description, descriptiongroup). Houses were defined as buildings with buildinguse_addresscount_residential > 0 and buildinguse_addresscount_total = 1. Hard-to-heat houses were identified by age (pre-1999) or poor construction material, using metadata-driven column selection. Folium was used for mapping, with datetime columns converted to string for compatibility. - Final outcome: 35,348 houses in Exeter were identified as likely hard to heat, and their distribution was visualized on an interactive map. The process involved careful stepwise filtering and use of metadata to ensure accurate results. - Overall reasoning: Each agent and tool was chosen based on its system prompt and capabilities. Parameters and columns were selected based on metadata and explicit instructions in the prompts. Errors in code execution were handled by adjusting variable names and data serialization as required by the tools.